Exercise7

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```r
library(dplyr) # A Grammar of Data Manipulation
library(ggplot2) # Create Elegant Data Visualisations Using the Grammar of Graphics
library(kableExtra) # Construct Complex Table with 'kable' and Pipe Syntax
library(mlogit) # Multinomial Logit Models
```

```r
data("Heating",
     package = "mlogit")
```

```r
H <- mlogit.data(Heating,
                 shape = "wide",
                 choice = "depvar",
                 varying = c(3:12))
```

1. Restate the blue bus-red bus situation as a nested logit model. What are the

marginal and conditional probabilities of this model?

The nested model organizes choices hierarchically. At the top level, individuals choose between two nests - either Car or Bus Nest. The Bus Nest contains the correlated alternatives: Blue Bus (bb) and Red Bus (rb). The dissimilarity parameter (λ) quantifies the degree of correlation within the bus nest. If λ tends to 1, it would indicate the correlation between the variables within the nests is zero.

First, define the aggregates utilities of the bus alternatives: $e^{IV_{bus}}$

Then the probabilities of choosing car:

$$P_{car} = \frac{e^{V_{car}}}{e^{V_{car}} + e^{IV_{bus}}}$$

The probabilities of choosing car:

$$P_{bus} = \frac{e^{IV_{bus}}}{e^{V_{car}} + e^{IV_{bus}}}$$

Conditional Probabilities

Probability of Blue Bus in the Bus Nest:

$$P_{bb|bus} = \frac{e^{V_{bb}/\lambda}}{e^{V_{bb}/\lambda} + e^{V_{rb}/\lambda}}$$

Probability of Red Bus in the Bus Nest:

$$P_{rb|bus} = \frac{e^{V_{rb}/\lambda}}{e^{V_{bb}/\lambda} + e^{V_{rb}/\lambda}}$$

Mariginal Probabilities

For the blue bus:

$$P_{bb} = P_{bus} * P_{bb|bus}$$

For the red bus:

$$P_{rb} = P_{bus} * P_{rb|bus}$$

2. Usemodel `nl2` in this chapter and calculate the direct-point elasticity at the mean

values of the variables, for an increase in the installation costs of Gas Central systems.

```
nl2 <- mlogit(depvar ~ ic + oc, H,
              nests = list(room = c( 'er', 'gr'), central = c('ec', 'gc', 'hp')),
              # this will set all lambdas to be the same
              un.nest.el = TRUE,
              steptol = 1e-12)
summary(nl2)
```

```
##
## Call:
## mlogit(formula = depvar ~ ic + oc, data = H, nests = list(room = c("er",
##     "gr"), central = c("ec", "gc", "hp")), un.nest.el = TRUE,
##     steptol = 1e-12)
##
## Frequencies of alternatives:choice
##       ec       er       gc       gr       hp
## 0.071111 0.093333 0.636667 0.143333 0.055556
##
## bfgs method
## 19 iterations, 0h:0m:0s
## g'(-H)^-1g = 0.0443
## successive function values within tolerance limits
##
## Coefficients :
##                   Estimate  Std. Error  z-value  Pr(>|z|)
## (Intercept):er -1.1290e+00  7.8672e-02 -14.3508 < 2.2e-16 ***
## (Intercept):gc -1.0474e-02  1.5358e-02  -0.6820 0.4952521
## (Intercept):gr -1.1817e+00  8.0048e-02 -14.7628 < 2.2e-16 ***
## (Intercept):hp -4.5459e-02  1.5516e-02  -2.9298 0.0033921 **
## ic             -8.3899e-05  2.5826e-05  -3.2486 0.0011598 **
## oc             -2.2841e-04  5.9509e-05  -3.8383 0.0001239 ***
## iv              2.7374e-02  3.2338e-03   8.4649 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Log-Likelihood: -1003.5
## McFadden R^2:  0.018329
## Likelihood ratio test : chisq = 37.473 (p.value = 3.6549e-08)
```

```
X_mean <- model.matrix(nl2)[1:5,]
alt <- index(H)$alt[1:5]
```

```
mean_ic <- H %>%
  group_by(alt) %>%
```

```r
  summarize(ic = mean(ic)) %>%
  arrange(alt)

mean_oc <- H %>%
  group_by(alt) %>%
  summarize(oc = mean(oc)) %>%
  arrange(alt)
```

```r
X_mean[,5] <- mean_ic$ic
X_mean[,6] <- mean_oc$oc
```

```r
# Electric central
exp_V_ec <-
  exp((X_mean[alt == c("ec"), "oc"] * coef(nl2)["oc"] +
         X_mean[alt == c("ec"), "ic"] * coef(nl2)["ic"])
               / coef(nl2)["iv"])
# Electric room
exp_V_er <-
  exp((coef(nl2)["(Intercept):er"] +
         X_mean[alt == c("er"), "oc"] * coef(nl2)["oc"] +
         X_mean[alt == c("er"), "ic"] * coef(nl2)["ic"])
               / coef(nl2)["iv"])
# Gas central
exp_V_gc <-
  exp((coef(nl2)["(Intercept):gc"] +
         X_mean[alt == c("gc"), "oc"] * coef(nl2)["oc"] +
         X_mean[alt == c("gc"), "ic"] * coef(nl2)["ic"])
               / coef(nl2)["iv"])

# Gas room
exp_V_gr <-
  exp((coef(nl2)["(Intercept):gr"] +
         X_mean[alt == c("gr"), "oc"] * coef(nl2)["oc"] +
         X_mean[alt == c("gr"), "ic"] * coef(nl2)["ic"])
               / coef(nl2)["iv"])

# Heat pump
exp_V_hp <-
  exp((coef(nl2)["(Intercept):hp"] +
         X_mean[alt == c("hp"), "oc"] * coef(nl2)["oc"] +
         X_mean[alt == c("hp"), "ic"] * coef(nl2)["ic"])
               / coef(nl2)["iv"])
```

```r
# Conditional probabilities of systems within the central nest
cp_c <- data.frame(ec = exp_V_ec / (exp_V_ec + exp_V_gc + exp_V_hp),
                      gc = exp_V_gc / (exp_V_ec + exp_V_gc + exp_V_hp),
                      hp = exp_V_hp / (exp_V_ec + exp_V_gc + exp_V_hp))

# Conditional probabilities of systems within the room nest
cp_r <- data.frame(er = exp_V_er / (exp_V_er + exp_V_gr),
                      gr = exp_V_gr / (exp_V_er + exp_V_gr))
```

```r
#After removing ec
mp <- data.frame(central = exp(coef(nl2)["iv"] * log(exp_V_ec + exp_V_gc + exp_V_hp))
                      / (exp(coef(nl2)["iv"] * log(exp_V_ec + exp_V_gc + exp_V_hp)) +
```

```r
                                exp((coef(nl2)["iv"] * log(exp_V_er + exp_V_gr)))),
                         room = exp(coef(nl2)["iv"] * log(exp_V_er + exp_V_gr)) /
                           (exp(coef(nl2)["iv"] * log(exp_V_gc + exp_V_hp)) +
                              exp((coef(nl2)["iv"] * log(exp_V_er + exp_V_gr)))))
```

```r
nlp <- data.frame(system = c("ec", "er", "gc", "gr", "hp"),
                  # Conditional probability
                  cp = c(cp_c$ec, cp_r$er, cp_c$gc, cp_r$gr, cp_c$hp),
                  # Marginal probability
                  mp = c(mp$central, mp$room, mp$central, mp$room, mp$central),
                  beta_ic = c(as.numeric(nl2$coefficients["ic"])),
                  beta_oc = c(as.numeric(nl2$coefficients["oc"])),
                  lambda = c(1 - as.numeric(nl2$coefficients["iv"]))) %>%
  # Joint probability
  mutate(p = cp * mp)
```

```r
nlp <- cbind(nlp, X_mean[,5:6]) %>%
  # Increase installation cost 1%
  mutate(ic_1pct = if_else(nlp$system == "gc", 1.01 * ic, ic))
```

```r
direct_elasticities <- nlp %>%
  transmute(DEM = ((1 - mp)    +    (1 - cp) * (1 - lambda)/lambda)    *    beta_ic * ic)

direct_elasticities
```

```
##           DEM
## 1 -0.01814786
## 2 -0.06446512
## 3 -0.01570029
## 4 -0.05979788
## 5 -0.02306906
```

3. Use model `nl2` in this chapter and calculate the cross-point elasticity at the mean values of the variables, for a 1% increase in the operation costs of Gas Central

systems.

```r
elasticities <- nlp %>%
  transmute(CEM_ec = -(mp    +  (1 - lambda)/lambda * cp)    *    beta_ic * mean_ic$ic[mean_ic == "ec"]
            CEM_er = -(mp    +  (1 - lambda)/lambda * cp)    *    beta_ic * mean_ic$ic[mean_ic == "er"]
            CEM_gc = -(mp    +  (1 - lambda)/lambda * cp)    *    beta_ic * mean_ic$ic[mean_ic == "gc"]
            CEM_gr = -(mp    +  (1 - lambda)/lambda * cp)    *    beta_ic * mean_ic$ic[mean_ic == "gr"]
            CEM_hp = -(mp    +  (1 - lambda)/lambda * cp)    *    beta_ic * mean_ic$ic[mean_ic == "hp"]
  # Transmute so that each row is the elasticity due to changes to a system
  t()
```

```r
diag(elasticities) <- direct_elasticities$DEM
```

```r
elasticities
```

```
##                  1           2           3           4          5
## CEM_ec -0.01814786  0.01710267  0.05446052  0.01763471  0.05294864
## CEM_er  0.06321789 -0.06446512  0.06498776  0.02104351  0.06318363
## CEM_gc  0.04991152  0.01611293 -0.01570029  0.01661418  0.04988447
## CEM_gr  0.05922422  0.01911935  0.06088228 -0.05979788  0.05919213
```

```
## CEM_hp  0.06723697  0.02170610  0.06911936  0.02238135 -0.02306906
```

4. Re-estimate the nested logit model in this chapter, but change the nests to types of energy as follows: - Gas: gas central, gas room. - Electricity: electric central, electric room, heat pump. Use a single coefficient for the inclusive variables (i.e., set un.nest.el = TRUE). Are the results reasonable? Discuss.

```r
nl1 <- mlogit(depvar ~ oc + ic, H,
             nests = list(gas = c('gc', 'gr'),
                          electrical = c('ec', 'er', 'hp')),
              un.nest.el = TRUE,
             steptol = 1e-12)
summary(nl1)
```

```
##
## Call:
## mlogit(formula = depvar ~ oc + ic, data = H, nests = list(gas = c("gc",
##     "gr"), electrical = c("ec", "er", "hp")), un.nest.el = TRUE,
##     steptol = 1e-12)
##
## Frequencies of alternatives:choice
##       ec       er       gc       gr       hp
## 0.071111 0.093333 0.636667 0.143333 0.055556
##
## bfgs method
## 14 iterations, 0h:0m:0s
## g'(-H)^-1g = 5.36E-07
## gradient close to zero
##
## Coefficients :
##                  Estimate Std. Error z-value  Pr(>|z|)
## (Intercept):er  0.8391327  0.6751569  1.2429  0.213916
## (Intercept):gc  1.0431426  1.4889576  0.7006  0.483562
## (Intercept):gr -3.1044881  1.3014025 -2.3855  0.017056 *
## (Intercept):hp -2.4162407  0.8791472 -2.7484  0.005989 **
## oc             -0.0095729  0.0023864 -4.0114 6.035e-05 ***
## ic             -0.0030007  0.0010566 -2.8398  0.004514 **
## iv              2.9493296  1.5797005  1.8670  0.061899 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Log-Likelihood: -1005.3
## McFadden R^2:  0.016517
## Likelihood ratio test : chisq = 33.768 (p.value = 2.218e-07)
```

The re-estimated nested logit model groups alternatives by energy type with a single inclusive value coefficient. Both cost variables (`oc`, `ic`) are negative and statistically significant, aligning with expectations—higher costs reduce utility. However, as the iv parameter (2.949) exceeds 1, violating the nested logit model's theoretical requirement for consistency. This suggests improper nesting or misspecification.

Alternative-specific constants reveal lower baseline utility for gas room (`gr`) and heat pump (`hp`) compared to electric central (`ec`), but gas central (`gc`) and electric room (`er`) constants are insignificant. The model's poor fit (McFadden $R^2 = 0.017$) further questions its validity.